Movie Memory Experiment

Report 1: Descriptive and Inferential Statistics

The Standard Deviants

1. Introduction

Our study investigated whether disrupting natural event boundaries during video viewing effect visual recognition memory. Participants watched YouTube Shorts clips under two conditions: **Natural Boundary (NB)**, where clips ended at a natural scene transition, and **Abrupt Boundary (AB)**, where clips were cut 1–5 seconds before the boundary. After viewing, participants completed a two-alternative forced-choice (2AFC) recognition test, identifying target frames (seen) from lure frames (unseen). Targets were drawn from either **before the boundary (BB)** or the **event middle (EM)** frames, allowing us to test whether any memory impairment is specific to boundary-adjacent frames or due to whole event.

2. Background

Human memory does not encode all moments equally. According to **Event Segmentation Theory** (EST; Zacks et al., 2007), the brain continuously builds a working model of the current situation and generates predictions about what will happen next. When incoming information violates these predictions a **prediction error** is generated, signalling the start of a new event. This triggers a model update that strengthens memory encoding for boundary-adjacent content, while making prior event information temporarily less accessible (Radvansky & Zacks, 2017). The result is the **boundary advantage**: recognition memory is better for frames near event boundaries than for frames from the event middle (Swallow, Zacks, & Abrams, 2009). By artificially cutting a clip before its natural boundary, the AB condition prevents the normal model-update process from completing, which should weaken the boundary-driven encoding boost and reduce memory for the disrupted content.

3. Dataset Description

A total of **185 participants** from IIIT Hyderabad community and are randomly assigned to one of two between-subjects conditions. In the **Natural Boundary (NB)** condition, participants watched video clips that ended at a natural event boundary. In the **Abrupt Boundary (AB)** condition, clips were cut 1–5 seconds before the natural boundary.

Each participant watched 45 video clips during encoding 40 unique YouTube Shorts and 5 repeated clips serving as **vigilance checks** (spacebar press on recognition). After encoding, participants completed a 2AFC recognition test for each unique clip, choosing between a **target** (seen) and a **lure** frame (unseen). Targets were either **BB** or **EM**. After each decision, participants rated confidence on a 1–5 scale. The measured variables are in Table 1.

Table 1. Measured Variables

Variable	**Type**	**Description**
Accuracy	Binary	Correctly identified target (1) or incorrect (0)
RT	Cont. (s)	Time from stimulus display to key press
Confidence	Ordinal	Self-rated certainty (1 = low, 5 = high)
Frame Type	Categorical	BB (near boundary) or EM (event middle)
Condition	Categorical	NB (natural boundary) or AB (abrupt boundary)
Enc. Time	Cont. (min)	Total time spent watching videos during encoding

4. Methods

4.1 Participant Demographics

A total of **185 participants** were recorded, of whom **171** had usable behavioural data files. Of these, **163** had valid demographic records; the remaining **22** were missing demographic information and are included in all core behavioural analyses but excluded from demographic-specific tests (H6). Both groups were well-matched on all demographic variables; a breakdown is given in Table 2.

Table 2. Participant Demographics by Condition

Variable	**Category**	**AB**	**NB**
Age	Range (years)	19–34	
	Mean (SD)	22.3 (2.3)	22.3 (2.4)
Gender	Male	55	58
	Female	27	23
Handedness	Right-handed	79	75
	Left-handed	3	6
Vision	Normal	44	40
	Corrected	38	39
	Uncorrected	0	2

4.2 Data Cleaning

Incomplete data. The file for participant sub42 (NB) contained only demographic output with no recognition trial data. Participant sub20 data was unavailable. Both were excluded from all analyses.

Excessive encoding time. The five repeated clips account for 5.6 minutes of total encoding time; assuming 25 seconds per repeated-clip recognition, participants spending more than 27.05 minutes in total were deemed inattentive. Four participants exceeded this threshold and were excluded (Table 3).

Table 3. Participants Excluded for Excessive Encoding Time

Participant	**Condition**	**Enc. Time (min)**
sub32	AB	27.33
sub36	AB	27.77
sub151	NB	27.16
sub161	NB	28.00

Response time outliers. Trials with RT < 0.2 s or RT > 60 s were excluded. One outlier was identified: sub28 (AB, Movie 8, RT = 71.09 s).

Response time cleaning. In some of the analysis we used the distributions and applied data cleaning. Trials with response times (RT) greater and lesser than $Mean + -4 \times SD$ were removed to minimize the impact of inattentiveness. This resulted in the removal of 54 trials (0.8% of total data).

Condition re-classification. Participant sub157 was labelled AB in the filename, but the video path column confirmed NB stimuli were viewed. Sub157 was re-classified to NB.

Missing demographic data. Twenty-two participants who did not fill the demographic form are retained in all core analyses but excluded from demographic balance tests (H6).

After all exclusions, the **final analysed sample comprised 166 participants** (AB: $n = 78$; NB: $n = 88$).

4.3 Hypotheses

Based on Event Segmentation Theory, we came up with six hypotheses.

H1 — Effect of Condition

IV	Viewing condition (NB vs. AB) — between subjects
DV	Mean recognition accuracy, mean RT (s)
H_0	No difference in accuracy or RT between NB and AB
H_A	Participants in the NB condition will show higher overall recognition accuracy and faster response times than participants in the AB condition.
Direction	One-tailed (NB > AB accuracy; NB < AB RT)
Test	Mann-Whitney U test & trial-level Regression

H2 — Effect Larger for BB Frames

IV	Viewing condition (NB vs. AB) — between subjects
DV	Mean recognition accuracy for BB frames
H_0	No difference in BB accuracy between conditions
H_A	Recognition accuracy will be greater for before-boundary (BB) frames than for event-middle (EM) frames.Reflecting NB advantage
Direction	One-tailed (Accuracy : BB > EM)
Test	Independent-samples t-test

H3 — No Condition Effect for EM Frames

IV	Viewing condition (NB vs. AB) — between subjects
DV	Mean recognition accuracy for EM frames
H_0	Recognition accuracy for EM frames will not differ significantly between the NB and AB groups.
H_A	Significant difference in EM accuracy
Direction	Two-tailed (EM frames are away from the manipulation point)
Test	Independent-samples t-test

H4 — Confidence Calibration

IV	Trial outcome (correct / incorrect) — within subjects
DV	Mean confidence rating (1–5 scale)
H_0	Confidence equal for correct and incorrect trials within each condition (AB and NB)
H_A	Within each condition (AB and NB), confidence ratings will be higher for correctly recognised trials than for incorrectly recognised trials
Direction	One-tailed
Test	Shapiro-Wilk (normality) → Paired t-test (if normal) or Wilcoxon Signed-Rank (if non-normal), applied separately per condition

H5 — Confidence by Frame Type

IV	Frame type (BB vs. EM) — within subjects
DV	Mean confidence rating (1–5 scale)
H_0	Confidence equal for BB and EM trials
H_A	Confidence ratings will be higher for BB trials than for EM trials.
Direction	One-tailed
Test	Paired-samples t-test

H6 — Demographic Balance	
IV	Viewing condition (NB vs. AB) — between subjects
DV	Age; gender; handedness; vision status
H_0	The NB and AB groups will not differ significantly on age, gender distribution, handedness distribution, or vision status.
H_A	Demographic distributions differ significantly
Direction	Two-tailed
Test	Independent t-test (age); χ^2 test (categorical variables)

4.4 Statistical Analysis

Normality of distributions was verified using the Shapiro-Wilk test and homogeneity of variance using Levene's test before each inferential test. Because the data violated normality for two conditions for H1, we used **Mann-Whitney U test**. For H4, where the NB group's difference scores were non-normal, the Wilcoxon Signed-Rank test was substituted for the paired t-test. For H6 age, where both group distributions were non-normal, the Mann-Whitney U test was used. The significance threshold was $\alpha = 0.05$ throughout. Also implemented regression models in H1 & H5 to evaluate predictive effects without losing trial-level variability.

5. Results

5.1 H1 — Main Effect of Condition

Prediction: NB participants would show higher accuracy and faster response times than AB participants.

5.1.1 Accuracy

We started with checking normality of the data through The Shapiro-Wilk test which indicated that participant accuracy was significantly non-normal for both groups ($p < .05$). As the groups are not related we implemented the non-parametric **Mann-Whitney U Test**.

Participants in the Natural Boundary (NB) group recognised significantly more targets ($Mdn = 87.5\%$, $M = 87.4\%$) than those in the Abrupt Boundary (AB) group ($Mdn = 85.0\%$, $M = 84.0\%$). The Mann-Whitney test confirmed this difference was statistically significant ($U = 2550.5, p = .0042$). We also conducted a trial-level **Logistic Regression** predicting accuracy. The model confirmed that condition is a strong independent predictor of accuracy; being in the NB condition significantly increased the log-odds of a correct response ($B = 0.280, p < .001$). This comprehensively confirms that following natural event boundaries leads to a significant increase in visual memory recognition.

5.1.2 Response Time

As discussed in the data cleaning trials with response times outside $Mean + 4 \times SD$ are excluded to minimize noise.

Shapiro-Wilk test confirmed that response times were non-normal and as groups are not-related similarly implemented **Mann-Whitney U Test**. Also performed trial-level **OLS Linear Regression** as response time is a continuous numerical variable for evaluating RT by condition.

Contrary to our prediction, both group medians (which provide a good measure alongside means) indicated that NB participants ($Mdn = 5.55$ s, $M = 5.75$ s) were **not** faster than AB participants ($Mdn = 5.36$ s, $M = 5.59$ s) but were slightly slower on average. The difference was not statistically significant ($U = 3246.0$, $p = .548$). The trial-level OLS LR confirmed that experimental condition was not a significant predictor of response time ($B = 0.162$, $p = .081$).

H1b (Response Time): Since $p = .548 > \alpha = .05$, we **fail to reject the null hypothesis**. There is insufficient evidence to support the alternative hypothesis (H_A: NB participants respond faster than AB participants).

plots/H1_accuracy_RT.png

Figure 5.1 — H1: Mean recognition accuracy and mean response time by condition (NB vs. AB).

5.2 H2 — Recognition Accuracy for Pre-Boundary Frames

Prediction: NB participants would show higher accuracy for before-boundary (BB) frames than AB participants.

Rationale: The boundary advantage predicts that frames just before a natural boundary benefit from heightened perceptual and attentional processing. In the AB condition, the artificial early cut eliminates this encoding boost for BB frames.

Normality tests showed non-normal distributions, but sample sizes were sufficient. Variance homogeneity was confirmed.

Table 5.1. Mean BB-frame recognition accuracy

Group	M	SD	n
AB	0.822	0.105	78
NB	0.857	0.097	88

NB participants ($M = 0.857$) achieved significantly higher BB-frame accuracy than AB participants ($M = 0.822$), $t(164) = -2.195$, $p = .030$, indicating that the natural encoding boost for boundary-adjacent frames is weakened when the boundary is artificially cut off.

H2: Since $t(164) = -2.195$ and $p = .030 < \alpha = .05$, we **reject the null hypothesis** and **support the alternative hypothesis** (H_A: NB participants show higher BB-frame accuracy than AB participants).

5.3 H3 — Recognition Accuracy for Event-Middle Frames

Prediction: EM frame accuracy would not differ significantly between conditions.

Rationale: If the boundary effect only strengthens memory for moments immediately surrounding the boundary, frames from the event middle should be encoded similarly regardless of condition.

Table 5.2. Mean EM-frame recognition accuracy

Group	M	SD	n
AB	0.853	0.093	78
NB	0.884	0.079	88

Contrary to prediction, NB participants ($M = 0.884$) were also significantly more accurate on EM frames than AB participants ($M = 0.853$), $t(164) = -2.319$, $p = .022$. The accuracy drop in the AB group was uniform across both BB and EM frames, suggesting that an abrupt cut hurts memory for the whole clip.

H3: Since $t(164) = -2.319$ and $p = .022 < \alpha = .05$ at the raw level, we **fail to retain the null hypothesis** (H_0: no difference in EM-frame accuracy) in the uncorrected test.

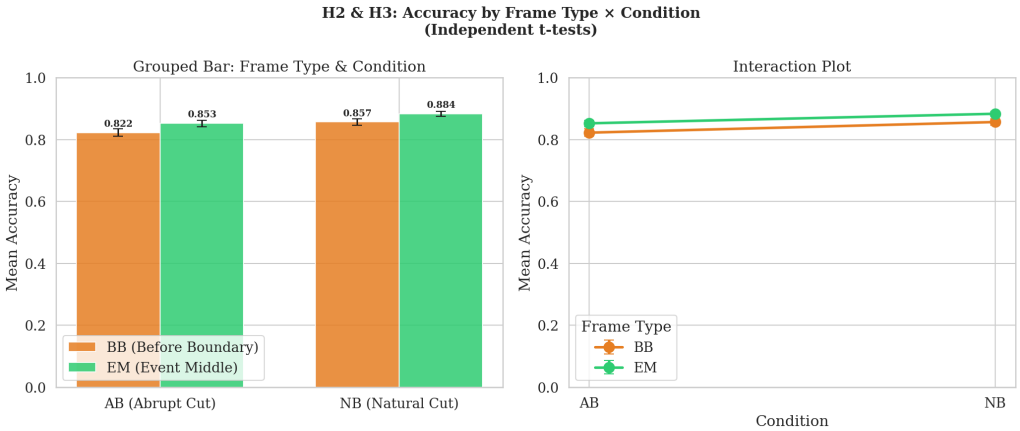

Figure 5.2 — H2 & H3: Mean recognition accuracy by frame type (BB, EM) and condition.

5.4 H4 — Confidence Calibration

Prediction: Within each condition (AB and NB), participants would report higher confidence on correct trials than on incorrect trials.

Rationale: If subjective confidence tracks genuine memory signal strength, it should be positively calibrated with accuracy. Analyzing this separately per condition confirms that metacognitive signals are robust regardless of boundary disruption.

One participant who made no errors across the session was excluded from this specific test.

We first evaluated the normality of the differences in confidence ratings using the Shapiro-Wilk test. For the AB group, the distribution was normal ($W = 0.975, p = .134$), leading to a paired t-test. For the NB group, the distribution was non-normal ($W = 0.969, p = .033$), requiring the Wilcoxon Signed-Rank test.

Table 5.3. H4 Results: Confidence Calibration by Condition

Group	Correct M	Incorr. M	Test	Stat	p	Effect Size
AB	4.201	3.362	t-test	$t = 12.562$	$< .001$	$d = 1.422$
NB	4.311	3.356	Wilcoxon	$W = 50.5$	$< .001$	$r = 0.971$

In both conditions, participants were significantly more confident when correct than when incorrect ($p < .001$), with the effect consistently in the predicted direction (**Correct** > **Incorrect**). Effect sizes were very large ($d = 1.422, r = 0.971$).

H4: Since $p < .001$ for both groups and the direction is correct, we **reject the null hypothesis** and **strongly support the alternative hypothesis**.

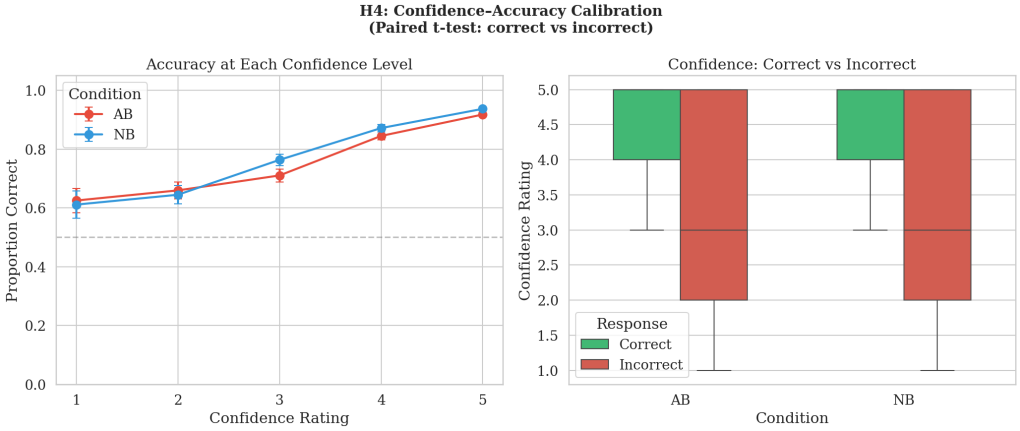

Figure 5.3 — H4: Mean confidence ratings for correct vs. incorrect trials by condition.

5.5 H5 — Confidence by Frame Type

Prediction: Participants would report higher confidence for BB frames than EM frames.

Rationale: If the boundary encoding advantage extends to subjective memory strength, participants should feel more certain about frames adjacent to the event boundary.

As discussed in the data cleaning trials with response times outside $Mean + 4 \times SD$ are excluded to minimize noise.

Examined confidence ratings (1–5) split by group. Normality tests were satisfied for both the AB group ($W = 0.978, p = .190$) and the NB group ($W = 0.982, p = .265$) so used **Paired t-tests**.

the results were mixed. For the **AB group** confidence was significantly higher for EM frames ($Mdn = 4.15, M = 4.12$) than for BB frames ($Mdn = 4.05, M = 4.03$). This difference was statistically significant ($t(77) = -3.004, p = .0036$). For the **NB group** there was no significant difference in confidence between EM frames ($Mdn = 4.30, M = 4.22$) and BB frames ($Mdn = 4.21, M = 4.20$; $p = .4097$).

We also did a trial-level **OLS Linear Regression** predicting confidence ($confidence \sim is_bb \times is_nb$). It confirmed being in the NB group ($B = 0.106, p = .008$) and viewing BB frames in the AB condition ($B_{is_bb} = -0.091, p = .025$) were significant predictors of confidence. NB participants showed higher baseline confidence overall, while AB condition participants showed a significant drop in confidence for BB frames. But the interaction term was not significant ($p = .222$).

H5 : We **fail to support the alternative hypothesis**. Because confidence for BB frames was either significantly *lower* than EM frames (AB group, $p = .0036$) or showed no significant difference (NB group, $p = .4097$). On the other hand we **reject the null hypothesis** as there was a difference in AB condition.

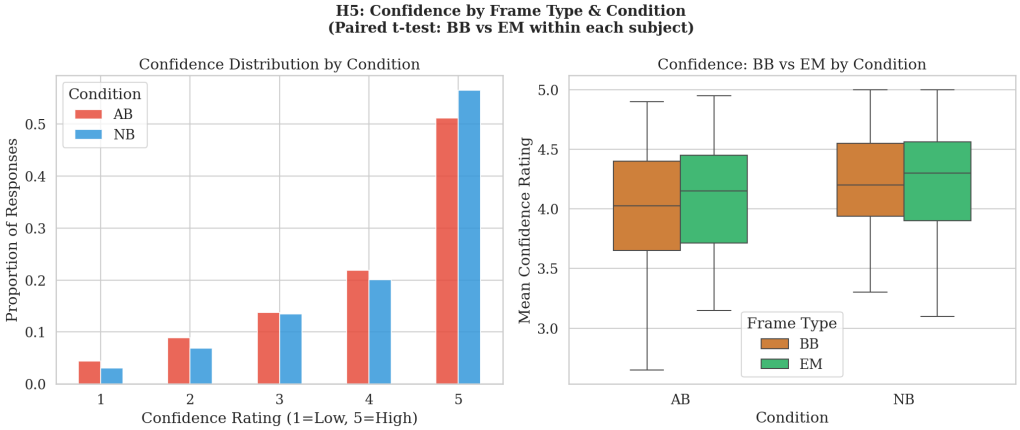

Figure 5.4 — H5: Mean confidence by frame type (BB vs. EM) and condition.

5.6 H6 — Demographic Equivalence

We first evaluated the normality of the age distribution using the Shapiro-Wilk test. Age was found to be significantly non-normal in both AB ($W = 0.855, p < .001$) and NB ($W = 0.929, p < .001$) groups. Consequently, an independent-samples Mann-Whitney U test was used for age. Categorical variables were tested using Chi-square tests of independence.

Table 5.5. Demographic Balance Checks

Variable	AB	NB	Test	Stat	p
Age	Mdn=22.3	Mdn=22.0	Mann-Whitney U	$U = 4391.5$	.564
Gender	51M, 27F	55M, 33F	Chi-square	$\chi^2(1) = 0.205$	.651
Handedness	71R, 7L	83R, 5L	Chi-square	$\chi^2(1) = 0.491$	.484
Vision	46N, 32C	45N, 43C	Chi-square	$\chi^2(2) = 2.161$	.339

There were no significant differences between groups on any demographic factor (all $p > .05$), confirming that random assignment produced balanced experimental groups.

H6: Since all $p > .05$, we **fail to reject the null hypothesis** and **support the alternative hypothesis** (groups are balanced).

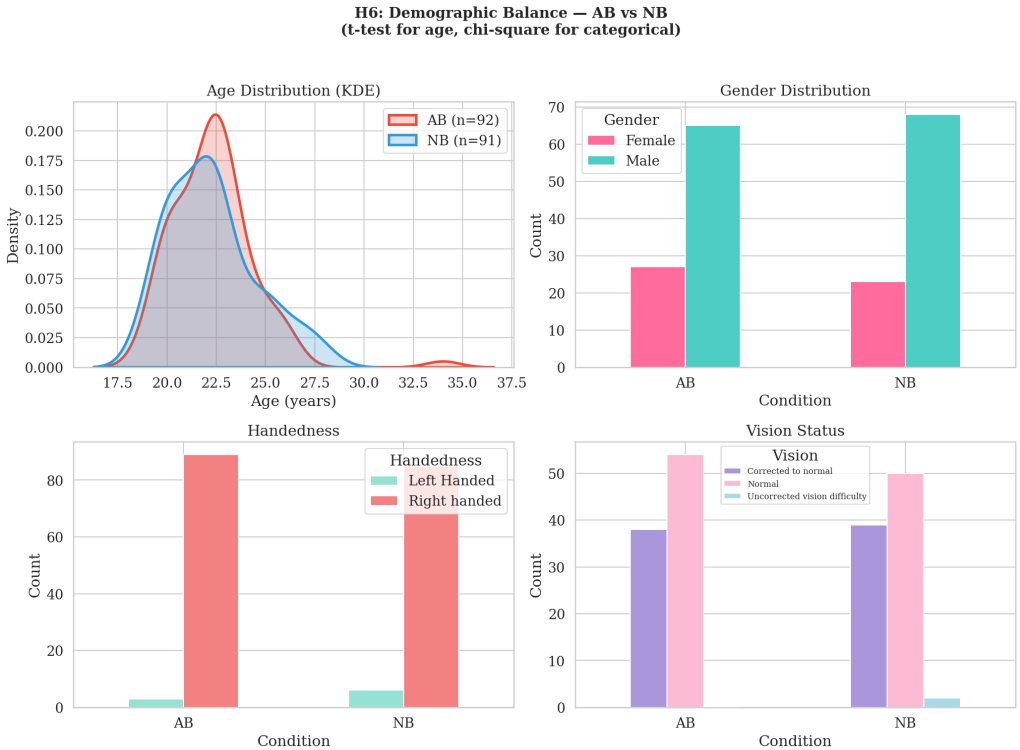

Figure 5.5 — H6: Age and gender distribution by condition.

Table 5.6. Participant-level means by condition

Variable	AB ($n = 78$)	NB ($n = 88$)
Accuracy (M, SD)	0.84 (0.08)	0.87 (0.07)
RT in s (M, SD)	5.57 (1.42)	5.75 (1.63)

6. Exploratory Analysis: Predicting Recognition Accuracy

While our previous tests looked at factors separately, we used a General Linear Model (Logistic Regression) to assess trials simultaneously ($N = 6,640$). This allowed us to test how confidence, experimental condition, and frame type (BB vs. EM) jointly predict accuracy.

Table 6.1. Logistic Regression Results for Recognition Accuracy

Predictor	Coef (B)	z	p	Odds Ratio
Intercept	-0.692	-5.99	$< .001$	0.50
Condition (NB)	0.183	2.51	.012	1.20
Frame Type (EM)	0.208	2.84	.004	1.23
Confidence	0.589	21.21	$< .001$	1.80

Confidence was the strongest independent predictor of accuracy; for every 1-point increase in subjective confidence, the odds of a correct answer increased by 80.2% ($OR = 1.80$). Both condition and frame type were also significant predictors, confirming that even when controlling for confidence, the natural-boundary advantage remains. Variance Inflation Factor (VIF) checks confirmed that all predictors were independent ($VIF \approx 1.0$), ensuring the model is stable.

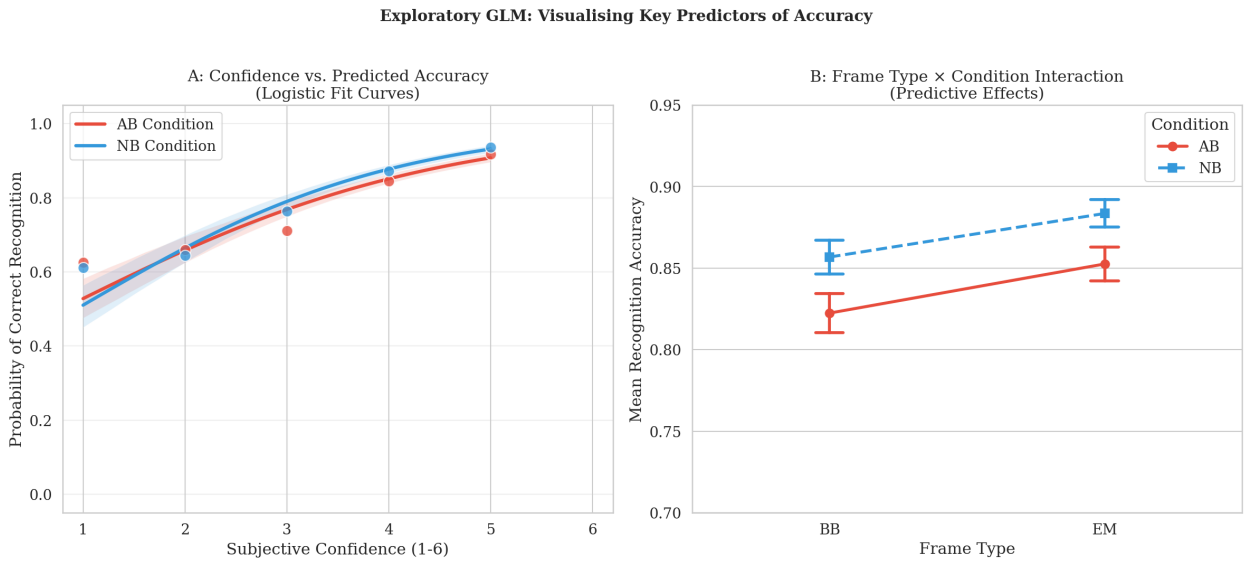

Figure 6.1: (A) Accuracy predicted by confidence; (B) Effect of video cuts and frame types.

7. Summary of Findings

- **Accuracy (H1a):** Cutting a clip early significantly reduced recognition accuracy, as confirmed by the non-parametric tests ($U = 2550.5, p = .0042$) and trial-level Logistic Regression models ($p < .001$). The effects were statistically significant for accuracy.
- **EM frames (H3):** The EM accuracy difference was also significant at the raw level ($p = .022$), contradicting the prediction of no difference. The boundary disruption appears to impair the whole event clip, not just boundary-adjacent frames.
- **RT (H1b):** No significant diff; NB participants were little *slower*, removing a speed–accuracy trade-off.
- **Confidence (H4):** Strongest result — participants were substantially more confident when correct than incorrect (AB: $d = 1.42$; NB: $r = 0.97$), demonstrating reliable metacognitive sensitivity.
- **Confidence by frame type (H5):** The AB group showed *lower* confidence for BB than EM frames — the opposite of the prediction. No effect in the NB group.
- **Demographics (H6):** Both groups were perfectly matched on all demographic factors (age, gender, handedness, and vision). No sample imbalances were found.
- **Exploratory Analysis (GLM):** Using a multi-factor model with $VIF \approx 1.0$, we found that confidence, condition, and frame type are all independent, significant drivers of recognition accuracy. Confidence is the most powerful predictor, yielding an 80.2% increase in the odds of accuracy per point.

8 References

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9. Contributions

- **Akash** — Data cleaning, Analysis for H1 and H5, report writing.
- **Varun** — Analysis for H4 and H6, Exploratory Analysis, report writing
- **Nikhilesh** — Analysis for H2 and H3, report writing.

Code Repository. https://github.com/Varun2345/BRSM-Project